

Signals & Systems (EE2008)

Final Exam

Date: May 22, 2024

Course Instructor(s)

Mr. Dr. S.M. Sajid

Mr. Asim Hussain Farooqi

Total Time (Hrs): 3
Total Marks: 100
Total Questions: 6

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Attempt all the questions.

CLO # 1: Express different types of signals and systems.

Q1: A continuous-time signal $x(t)$ is shown in Fig-1 below. Compute and label carefully each of the following signals: [20]

a) $x(2t + 1)$

b) $[x(t) + x(-t)]u(t)$

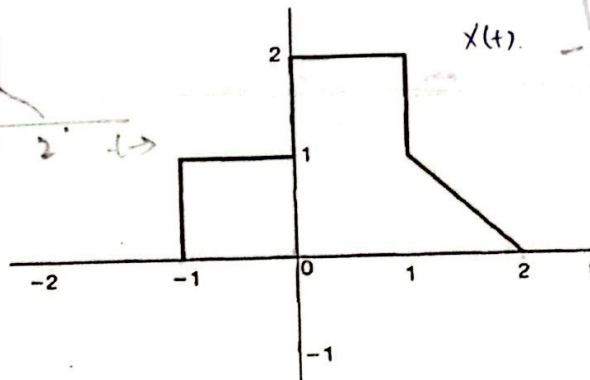
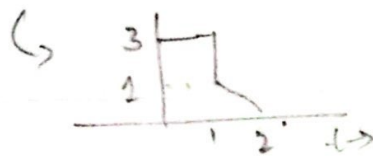
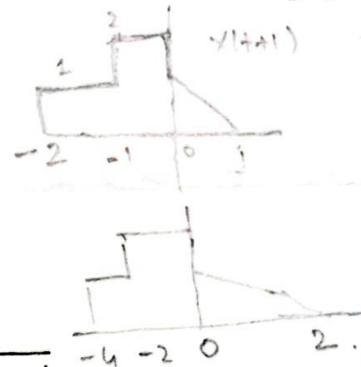
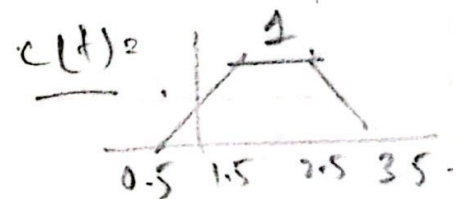
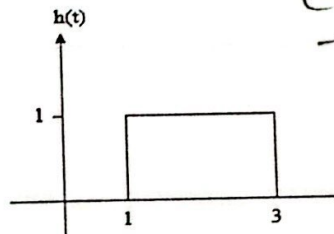
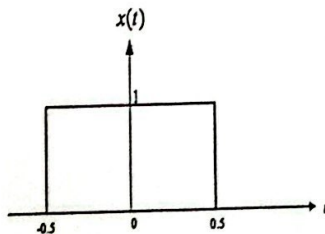


Fig-1



CLO # 2: Apply mathematical and graphical convolution techniques to analyze CT and DT systems.

Q2: Solve the convolution by graphical method, the signal $x(t)$ and $h(t)$ are given below. [20]



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$$y(t) = \begin{cases} t - 0.5 & 0.5 \leq t \leq 1.5 \\ 1 & 1.5 \leq t \leq 2.5 \\ -t + 3.5 & 2.5 \leq t \leq 3.5 \end{cases}$$